MSCI 391: Business Analytics Project- *Reflective Report*

Group E

Energy Consumption Modelling (OR2)

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Introduction & Background

Our project's aim was to act as consultants and analysts from a research institute, to inform Government policymakers regarding the UK's residential energy consumption and the transition to sustainable energy. We did this by gathering vast amounts of data on the UK's consumption of various energy sources and their average prices over time (mainly gas and electricity); then creating models to simulate and forecast demand for the next 5 years. These models were also used to conduct a top-down econometric analysis using Multiple Linear Regression, which required navigating complex macroeconomic data to create two distinct forecast scenarios to advise policymakers with.

To critically reflect on how our project went as a team, and to separate my individual roles and personal development, this essay will focus on two core themes: '***Teamwork'*** *and '**Personal Development'**.*

Overall as a **team** we judge the project a success, and the grades and feedback we received during the process confirm this- further details on how we managed this project will be discussed in section *'Theme 1. Teamwork'*. Generally, we had regular meetings during which we worked together efficiently and even completed parts of the project much earlier than expected. Our presentation also received highly constructive feedback, rather than getting 'grilled' by questions by the panel of clients, like we heard other groups experienced.

This project was also a personal success for my **personal development**. For context, I have worked previously with one of our teammates from this module on another project 2 years ago, however my personal circumstances and difficulties at the time led to poor contribution and fractured trust. This group project gave me an opportunity to implement reflections and changes I have made since then as well as build on them further, and to rebuild trust and 'redeem' myself.

These two themes are the focus of this reflective report. This also includes evaluation of our group dynamics, as well as my individual contribution. All the relevant stages of the project- including events that happened, decisions made and thought processes behind them- are explained in detail and analysed through the lens of these themes, exploring:

- what happened and why (sense-making),
- the impacts and consequences,
- the underlying causes,
- what I learned from the experiences (including linking to theoretical ideas),
- and actions I'll take to avoid similar mistakes or ensure repeat successes.

Theme 1. Teamwork

1.1 'Sense-Making' & Evaluation of 'What We Did'

From the beginning of the project, our group of four employed a flat, agile structure with no real 'leader' or project manager. However, at times control fluctuated, with some of us leveraging our individual skills more on certain tasks. No one had set 'roles'- we held meetings on average once a week where we subdivided tasks, but worked collaboratively in the same room for at least an hour, making holding each other accountable easier. If someone was absent, their work was easily covered since everyone oversaw each part of the project together- or the absentee would simply catch up by the next meeting.

We had no real hierarchy and roles were relatively fluid, however those with particular skills- such as with regression, data, research, designing presentations and documents, etc.- tended to gravitate to tasks that required those skills. For example I had a shared role in researching and collecting datasets to be used for the model, however the mathematical aspect of linear regression on Excel is not one of my strengths.

This co-working prevented the 'silo effect' common in group projects for university where students can tend to work separately, or even in corporate environments with fragmented employees. This meant our final report was written with one cohesive 'voice', rather than four disjointed parts written in very different styles/formats. It also removed the stress of single-point failures which could stall progress such as absent members.

One of my contributions to the group was immediately setting up and maintaining a shared Google Drive workspace with organised folders and documents where we could all work on the same report simultaneously, removing the friction of version control. Since we had all collected hundreds of pages of documents, powerpoints, drafts, lecture slides etc. throughout this module, it made sifting through Moodle pages and PDFs a lot quicker and easier which may have contributed to our efficiency as a team.

We took *some* inspiration from the PRINCE2 framework for project management to create a controlled work environment- however we soon realised this was not necessary for us, as the comfortable, informal working dynamic we naturally had as a team was working well (*more in section 1.2)*. Other frameworks such as SSM, CATWOE and the Rich Picture may have helped with understanding the scope of the problem, but that section felt like a formality compared to other, more important sections of the report and presentation such as the analysis or recommendations (*more in Theme 2).* The frameworks from the module such as TRIZ or PRINCE2 seemed more helpful for larger companies solving much more complex problems.

1.2 Critical Analysis: Relating Team Experiences to Theory

'Agile' Methodology

'Agile' project management rejects the traditional 'Waterfall' methodology of rigid, sequential planning and task allocation dictated by a project manager. An 'agile' team is a 'self-organising' one, where unresolved tasks get pulled by teammates from a shared backlog based on importance, urgency and personal skills, so work is allocated without the need for a hierarchy. We also labelled claimed sections with names to avoid confusion.

Our group's fluid role-taking perfectly portrays the efficiency of this philosophy. Instead of waiting for directions, team members instinctively claim tasks based on the project's needs and individual capacity, such as my decision to create the shared Drive and tackle SSM sections when no one else was choosing them. The centralised workspace effectively created a transparent task backlog, as well as an organised area for all documents resources regarding the project, which allowed us to bypass administrative and hierarchical bottlenecks.

'TRIZ': 'Equipotentiality' and 'Spheroidality'

Despite TRIZ and PRINCE2 not being massively useful for this particular project *(more on this in 1.3)*, retrospectively these frameworks can be used as a strong tool in managing bigger, more complex teams in the future.

TRIZ *(The Theory of Inventive Problem Solving)* identifies 40 specific 'inventive principles', or universal strategies for resolving contradictions. While this was not intended as we did not actively attempt to identify any internal contradictions, we strongly aligned with principles 12 (Equipotentiality) and 14 (Spheroidality). In business terms, these principles advocate for flattening hierarchies and empowering team members equally, and smoothing interactions by avoiding rigid boundaries and encouraging overlapping responsibilities. Operating under these principles ensured there was no single point of failure, and empowered every member to seamlessly absorb and share workloads without stalling the project's momentum.

'PRINCE2' & 'RACI Matrix'

PRINCE2 *(Projects IN Controlled Environments)* is a structured project management framework designed to maintain control and prevent project drift. A common tool related to this is the 'RACI Matrix', mapping who is *Responsible*, *Accountable*, *Consulted*, and *Informed* for every single task to prevent ambiguity.

However, a core principle of PRINCE2 is also to 'tailor to suit the environment'. Implementing such tools for our environment of a motivated, four-person university group would have introduced unnecessary bureaucracy or ‘over-management’, as we did not struggle with project drift. By tailoring our approach to avoid such rigid methodologies, we maintained agility- although operating too ‘lean’ this way carries scalability risks *(more in section 1.3).*

1.3 Learning Points, Solutions & Actions for Better Teamwork

A learning point from this project related to teamwork is that an organised, centralised workspace that all team members understand and can look at and work on together, reduces administrative friction and lays the foundations to enable agile ‘self-organisation’ of a team. Additionally, informal, flat structures (embodying the TRIZ principle of Equipotentiality) are effective for driving speed, cohesion and motivation in small groups.

However, critically evaluating this ‘RACI-free’ approach reveals some inherent limitations, especially regarding scalability. Relying on overlapping skills and unspoken trust may have succeeded for our four-person university project- however, if applied to a larger corporate consultancy engagement, a lack of defined and communicated *Responsibility* and *Accountability* enables significant project risks such as duplicated or missed deliverables.

These learning points and solutions can be improved and adapted for future professional environments in the following ways:

1. Firstly, I will continue to take initiative to establish easy-to-use collaborative workspaces on day one of any project, having seen firsthand how this accelerates a team through the initial ‘forming’ stage into ‘performing’.
2. Minimising governance and management tools like PRINCE2 or RACI is not always a good idea. Larger or more fragmented groups require a hybrid approach, combining the collaborative and agile working environment with formalised responsibility frameworks. This ensures that the team remains flexible, while keeping *Responsibility* and *Accountability* (and communication of these) clear as project complexity grows.

Ultimately the success of this fluid team dynamic was heavily built on mutual trust and individual reliability. My lack of reliability in a previous module created distrust and consequently disruption to the process- being in a group with the same person two years later subconsciously made me feel the need to ‘redeem’ myself and actively rebuild that trust, implementing reflections I have made since then to become a better team player.

This introduces the basis of my focus on personal development in Theme 2 below.

Theme 2. Personal Development

2.1 ‘Sense-Making’ & Evaluation of ‘My Contributions’

To fully contextualise my personal development, it is necessary to reflect on my starting point before this project. Personal difficulties during my second year of study resulted in a severe lack of contribution on my part to a similar group project, which understandably caused frustration at the time.

Entering this project in a group containing an individual I have let down before created some initial tension and put my trust and reliability under scrutiny. However, driven by the desire for ‘redemption’, rebuilding trust and getting good grades, I used this situation as a catalyst to implement reflections I have made since then, and adopt a proactive stance as a member of the group as soon as possible.

To show this initiative immediately, as soon as our group was formed I created and shared a Drive folder for this project. From my technical experience, Google’s workspace is the best for group projects like this where multiple people need live edit access to one shared document, as well as an organised library of resources and data relevant to the project. I leveraged my knowledge of other technical tools that greatly helped in our project, including:

- search tools and libraries of structured datasets. We would need lots of data for this project, and I took on a share of responsibility for researching, selecting and validating appropriate datasets based on assumptions, limitations and scope of the model
- Gemini Pro subscription, including ‘Nano Banana’ for AI image generation.

While we were all looking at the first ‘project outline’ assignment brief and assessing which sections to claim based on our skills and interests, I volunteered to take on the SSM section which other members seemed to be avoiding, and generated a surprisingly good AI ‘Rich Picture’ for that section.

Since that project outline until the final report, I had been working on the sections regarding those frameworks, which my peers viewed as a formality or ‘side-task’ compared to the actual model and analysis. I applied my technical knowledge once again in the presentation by designing a small animation which visually aided the slides on the Root Definition and CATWOE framework, communicating our ideas as clearly as possible to the client. This is confirmed by feedback from the presentation which I recorded and shared with my group, as I knew it would be helpful. Taking ownership of these ‘unwanted’ tasks and leveraging my knowledge and skills to improve the process for everyone, was underlyingly caused by an internal motivation for personal development with team skills, and to prove myself to my team and lead us to success.

2.2 Critical Analysis: Relating Personal Growth to Theory

'Systems Thinking' & Shifting Feedback Loops

'Systems Thinking' defines 'Reinforcing Loops' as loops of events that amplify change. Within the system of a team, consistent lack of contribution of one member creates a negative reinforcing loop, which ultimately results in frustration and worse output. Instead, in this project I proactively injected a positive loop as soon as possible by taking initiative to create a nice workspace for the team- rebuilding trust by creating psychological safety, leading to efficient teamwork.

Hard vs. Soft Problems (SSM)

Soft Systems Methodology (SSM) distinguishes between 'Hard' problems with clear and quantifiable solutions, and 'Soft' problems which are inherently human, qualitative and unpredictable. For me personally, this project possessed characteristics of both. While the core econometric analysis and regression is a textbook 'hard' problem, my personal challenges created a 'Soft' problem to overcome with my team. Ironically, it was the SSM concepts of learning and purposeful action that I used to navigate this 'messy', human situation.

'Forming, Storming, Norming, Performing'

These four stages of Bruce Tuckman's 'Stages of Group Development' can be used to map the trajectory of our group over time. Because of initial inter-personal tension, our group may have 'Formed' with an early 'Storming' phase of friction. However I quickly counteracted this and pushed for the 'Norming' and 'Performing' phases by establishing good habits early so we could begin 'Performing' well as soon as possible.

2.3 Learning Points, Solutions & Actions for Personal Improvement

A lot of my personal learning points and solutions will overlap with section 1.3. However, there are some key takeaways specifically regarding my personal development, which can also be applied to teamwork on future projects.

Primarily, *actions speak louder than words*. People generally appreciate when you take initiative, do things that make their job easier and solve their problems rather than just talk and plan, especially in a group environment with deadlines where you want to accelerate the group to the 'Performing' stage. This is also how trust is rebuilt- not just by apologising, but through visible action. Taking on ownership of organisational housekeeping and 'undesirable' administrative tasks are ways to build social capital in any workplace. For future professional roles, it would be a good idea to make an early habit of identifying ‘gaps’ in work that needs to be done, and either filling them in quickly to establish reliability, or solving the problem that caused that gap.

Furthermore, this project brought new light on the relationship between technical analytics and human variables. Previously, I viewed frameworks like SSM as mere academic formalities, compared to the ‘real’ work of econometric data modelling. However, the concepts of this tool helped with mapping our client’s messy reality, as well as navigating my own team’s dynamics. Managing ‘Soft’, interpersonal and structural problems is absolutely necessary to solve complex ‘Hard’ problems- technical proficiency can be easily undermined by poor communication or fractured trust. Going forward in my career, I plan to implement such frameworks to avoid the common analytical pitfall of going straight into the numbers before fully assessing the client’s messy reality.

3. Conclusion

The technical success of our energy consumption model and our group's deliverables was underpinned by the human elements of the project. Ultimately, this project was rather 'Soft' for a 'Hard' mathematical problem; meaning there were countless, unpredictable sociological /human variables that we could not control for, especially regarding energy prices. Because energy demand is not completely inelastic to price, predicting demand of electricity and gas is a difficult task bearing in mind the volatility of energy prices, especially in the current geopolitical climate.

As a result our recommendations were potentially rather generic- however in our defense, this is a much more complex issue of government, law and policy that has been analysed by experts globally for decades, with the additional looming threat of extreme volatility from climate change and geopolitics making this an essentially impossible task in the long term.

Reflection on Theme 1, this project proved that a rigid hierarchy with project managers dictating sequential task allocation is not always the default choice. Operating with agility and embracing a flat, 'equipotent' hierarchy can reduce certain frictions and help create a more homogenous, consistent product. The critical takeaway, however, is that this informal structure only succeeds when supported by a high degree of psychological safety. When responsibility and accountability are not naturally communicated and defined in a group, frameworks like RACI exist to be implemented to solve such structural issues.

Finally, regarding Theme 2, this project served as a successful 'proving ground' for my social and academic personal development. Overcoming historical tensions required visible action and willingness to take initiative and accountability, in order to regain trust. Transforming past failures from that difficult second year into general successes this year and in this module was perhaps one of my most significant achievements of my university experience. This module has given me blueprints to carry from this experience, to help master interpersonal dynamics and solve any complex business problem, individually or as a team.